

Dinkelbach-Guided Deep Reinforcement Learning for Secure Communication in UAV-Aided MEC Networks

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Abstract—Unmanned aerial vehicle-aided (UAV-aided) mobile edge computing (MEC) network can greatly reduce the data growth pressure of Internet of Things (IoT) and expand the wireless communication coverage. However, there is a risk of eavesdropping on the offloading information of terminal users (TUs) because of UAV light-of-sight (LoS) transmission. In this paper, we propose a Dinkelbach-guided deep reinforcement learning (DRL) scheme for secure communication in the UAV-aided MEC network. Specifically, the security calculating efficiency of the network is maximized by optimizing offloading decision and resource allocation under the condition of the data queue stability and minimum calculating requirement. The problem is intractable due to the fractional structure and binary constraint. Firstly, we deal with the fractional structure by taking advantage of Dinkelbach optimization. Then, offloading decision is generated based on DRL and the resource is allocated by successive convex approximation (SCA). Simulation results show that the proposed Dinkelbach-guided DRL scheme efficiently improves the security calculating efficiency of the network.

Index Terms—UAV, MEC, deep reinforcement learning, secure communication, security calculating efficiency

I. INTRODUCTION

With the rapid development of Internet of Things (IoT), human beings have higher demand for material culture, which leads to a tremendous increase in IoT data traffic [1]- [2]. Mobile edge computing (MEC) is viewed as a promising technology to promote the evaluation of IoT real-time information processing for high speed and low energy consumption [3].

MEC allows terminal users (TUs) to offload the data information to calculate at the edge [4]- [7]. Malik *et al.* pointed out MEC can provide a high-speed and low-energy service with assisting TUs to calculate the task information [5]. Sun *et al.* studied the task allocation scheme for MEC-IoT networks to highlight response rate [6]. Wang *et al.* proposed a distributed optimization scheme for MEC networks to augment TUs' calculating capabilities [7].

To improve the quality and coverage of wireless communication, UAVs can be used to carry MEC server to deploy in computationally intensive places flexibly [8]- [11]. Han *et al.* proposed a rate splitting scheme for the UAV-aided MEC network to enhance the communication transmission quality [9]. Hu *et al.* studied the maximizing calculating efficiency scheme through optimizing UAV trajectory and resource management

alternatively [10]. Du *et al.* investigated an efficient method to enhance the performance of UAV-aided MEC networks with time division multiple access (TDMA) workflow [11].

However, there is a risk of eavesdropping on the offloading information of terminal users (TUs) because of UAV light-of-sight (LoS) transmission [12]- [14]. Lu *et al.* studied the secure calculating performance maximization scheme of the dual-UAV-MEC network by considering the calculating capability of TUs and UAV [12]. Zhou *et al.* proposed an efficient secure communication scheme for UAV-aided MEC networks through optimizing TUs' association and resource allocation. Lu *et al.* investigated NOMA-based UAV-MEC network's security calculating capacity maximization through optimizing communication and computation resource and UAV trajectory [14].

The above works almost take multiple iterations to obtain the optimizing solution with the high complexity. Deep reinforcement learning (DRL) is considered as an efficient method to improve the secure calculating performance for the networks [15]- [18]. Nath *et al.* proposed a DRL-based online calculating method to achieve the efficient performance for the MEC network [16]. Bi *et al.* proposed an effective online offloading algorithm for MEC networks, which takes the advantage of DRL [17]. Wu *et al.* investigated the DRL-based calculating algorithm to improve MEC network's communication quality [18].

However, there is little work to introduce DRL for secure communication in UAV-aided MEC networks, which is worth studying.

Therefore, in this paper, we propose a Dinkelbach-guided DRL scheme for secure communication in UAV-aided MEC networks. The security calculating efficiency of the UAV-aided MEC network is maximized by optimizing offloading decision and resource allocation, including local calculation and time allocation. Specifically, we consider a UAV-aided MEC network for secure communication, where TUs' data queue changes randomly per frame. TUs offload their data information to compute at Server UAV (SV). During the offloading transmission, Eavesdropping UAV (EV) eavesdrops the data information. Ground jammer (GJ) sends interfering information to thwart EV eavesdropping. Under the condition of data queue stability and basic calculating requirement, we formulate a problem to

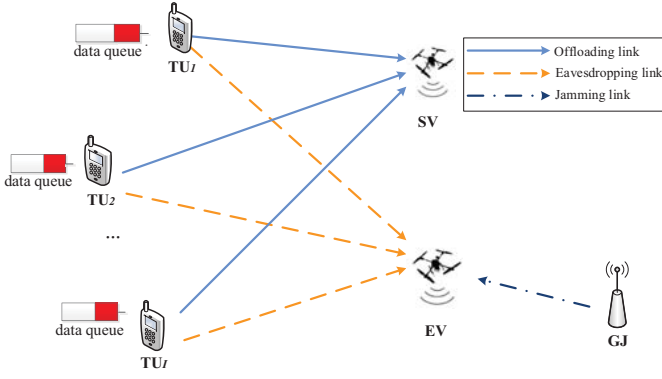


Fig. 1. The UAV-aided MEC Network.

maximize the security calculating efficiency of the UAV-aided MEC network by optimizing offloading decision and resource allocation, including local calculation and time allocation. The problem is intractable due to the fractional structure and binary constraint. Firstly, we deal with the fractional structure by taking advantage of Dinkelbach optimization. Then, we use DRL to generate offloading decision, where the deep neural network (DNN) learns the offloading decision from the replay memory, and the resource is allocated by SCA.

II. SYSTEM MODEL

Fig. 1 shows the UAV-aided MEC network for secure communication we consider, where I TUs' data queue changes randomly per frame. TUs offload their data information to compute at Server UAV (SV). During the offloading transmission, Eavesdropping UAV (EV) eavesdrops the data information. Ground jammer (GJ) sends the interfering information to thwart EV eavesdropping. We assume that SV can distinguish GJ's interfering information from the received signal without being affected by GJ's interfering information. However, EV belongs to the illegitimate network. GJ's interfering information will affect EV eavesdropping.

SV and EV perform calculating offloading task and eavesdropping task with the predetermined trajectory, respectively. Assume that the one frame n is δ_t , the locations of SV and EV in frame n are $w_s[n]$ and $w_e[n]$. Define the locations of TU_i , $i \in \{1, 2, \dots, I\}$ and GJ as w_i and w_j . In the UAV-aided MEC network, there are new data queue $A_i[n]$ arrives at TU_i per frame.

The distance between SV and TU_i , EV and GJ, EV and TU_i are $d_{i,s}[n] = \sqrt{\|w_s[n] - w_i\|^2}$, $d_{j,e}[n] = \sqrt{\|w_e[n] - w_j\|^2}$ and $d_{i,e}[n] = \sqrt{\|w_e[n] - w_i\|^2}$ in frame n . The channel of the UAV-aided MEC network is followed as Rician fading channel model. The channel coefficient between SV and TU_i , EV and

GJ, EV and TU_i are expressed as

$$h_{i,s}[n] = \sqrt{\frac{G}{G+1}} \sqrt{\frac{\beta_0}{d_{i,s}[n]^2}} e^{j\theta_{i,s}[n]} + \sqrt{\frac{1}{G+1}} \sqrt{\frac{\beta_0}{d_{i,s}[n]^2}} f_{i,s}[n], \quad (1)$$

$$h_{j,e}[n] = \sqrt{\frac{G}{G+1}} \sqrt{\frac{\beta_0}{d_{j,e}[n]^2}} e^{j\theta_{j,e}[n]} + \sqrt{\frac{1}{G+1}} \sqrt{\frac{\beta_0}{d_{j,e}[n]^2}} f_{j,e}[n], \quad (2)$$

$$h_{i,e}[n] = \sqrt{\frac{G}{G+1}} \sqrt{\frac{\beta_0}{d_{i,e}[n]^2}} e^{j\theta_{i,e}[n]} + \sqrt{\frac{1}{G+1}} \sqrt{\frac{\beta_0}{d_{i,e}[n]^2}} f_{i,e}[n], \quad (3)$$

where β_0 is the power gain at $1m$, G is Rician factor, $\theta_{i,s}[n]$, $\theta_{j,e}[n]$ and $\theta_{i,e}[n]$ are the LoS channel's phase of SV and TU_i , EV and GJ, EV and TU_i , respectively. $f_{i,s}[n]$, $f_{j,e}[n]$ and $f_{i,e}[n]$ follows $CN(0, 1)$.

III. PROBLEM FORMULATION

TUs adopts TDMA work mode to offload the data information. Denote $\tau_i[n]$, $i \in \{1, 2, \dots, I\}$ as TU_i time factor in frame n , which follows the constraint,

$$\sum_{i=1}^I \tau_i[n] \leq 1, \quad (4a)$$

$$0 \leq \tau_i[n] \leq 1. \quad (4b)$$

A. Communication Model

SV can distinguish GJ's interfering information from the received signal without being affected by GJ's interfering information. However, EV belongs to the illegitimate network. GJ's interfering information will affect EV eavesdropping.

The data offloading rate (bps/Hz) at SV and the eavesdropping rate (bps/Hz) at EV in frame n is

$$R_{i,s}[n] = \tau_i[n] \log_2 \left(1 + \frac{|h_{i,s}[n]|^2 p_i[n]}{\sigma^2} \right), \quad (5)$$

$$R_{i,e}[n] = \tau_i[n] \log_2 \left(1 + \frac{|h_{i,e}[n]|^2 p_i[n]}{\sigma^2 + |h_{j,e}[n]|^2 P_j} \right), \quad (6)$$

where $p_i[n]$ and P_j are TU_i and GJ's transmit power, σ^2 is the Gaussian noise.

Therefore, the security calculating rate is obtained as

$$R_{i,sec}[n] = R_{i,s}[n] - R_{i,e}[n]. \quad (7)$$

B. Calculation Model

In this UAV-aided MEC network for secure communication, TUs use the binary offloading strategy to calculate their data queue. Denote $r_i[n]$ as TU_{*i*}'s offloading decision, which satisfies

$$r_i[n] = \begin{cases} 0, & \text{local calculation,} \\ 1, & \text{offloading calculation.} \end{cases} \quad (8)$$

We know that the calculating abilities of TUs and SV are limited. In local calculation, TU_{*i*}'s local calculating bits $l_{loc,i}[n]$ should satisfy

$$c_i l_{loc,i}[n] \leq F_i^{\max} \delta_t, \quad (9)$$

where c_i is CPU cycles for TU_{*i*} calculate one bit, and $F_i^{\max}$ is TU_{*i*} maximum CPU frequency.

Similarly, in offloading calculation, the calculating bits that SV assists TU_{*i*} to calculate should satisfy

$$B c_s R_{i,sec}[n] \delta_t \leq \tau_i[n] \delta_t F_s^{\max}, \quad (10)$$

where c_s is CPU cycles for SV calculate one bit, B is bandwidth, and $F_s^{\max}$ is SV maximum CPU frequency.

The energy consumed in local calculation and offloading calculation are given by

$$E_{loc,i}[n] = \frac{(c_i l_{loc,i}[n])^3 k_i}{\delta_t^2}, \quad (11)$$

$$E_{off,i}[n] = \tau_i[n] p_i[n] \delta_t, \quad (12)$$

where k_i is TU_{*i*}'s capacitance coefficient.

Then, we can obtain the sum security calculating bits and energy consumption of TU_{*i*},

$$l_i[n] = r_i[n] B \delta_t R_{i,sec}[n] + (1 - r_i[n]) l_{loc,i}[n], \quad (13)$$

$$E_i[n] = r_i[n] E_{off,i}[n] + (1 - r_i[n]) E_{loc,i}[n]. \quad (14)$$

Assuming that $q_i[n]$ is TU_{*i*}'s data queue length at the beginning of the time frame n . TU_{*i*}'s next queue is

$$q_i[n+1] = q_i[n] - l_i[n] + A_k[n]. \quad (15)$$

In the UAV-aided MEC network, TUs have the basic calculating requirement q_{min} in every frame, which can balance each TUs' data will be calculated. Thus, we have

$$l_i[n] \geq q_{min}. \quad (16)$$

We know that TU_{*i*} cannot be calculated with more data than it has,

$$l_i[n] \leq q_i[n]. \quad (17)$$

Denote Q_{max} as the maximum data storage of each TU. The data queue stability constraint is as follows

$$q_i[n] \leq Q_{max}. \quad (18)$$

C. Problem Formulation

Due to $A_i[n]$ is unknown, we maximize the security calculating efficiency of the UAV-aided MEC network in every frame $\eta[n] = \sum_{i=1}^I \left(\frac{l_i[n]}{E_i[n]} \right)$ to achieve online offloading and resource optimization. We formulate the optimization problem as

$$(P1) : \max_{\{r_i[n], \tau_i[n], l_{loc,i}[n]\}} \eta[n] \quad (19)$$

$$s.t. (4), (8), (9), (10), (16), (17), (18).$$

The problem (P1) is difficult to solve due to the fractional structure and binary constraint.

IV. PROBLEM SOLUTION

To efficiently deal with the fractional structure of (P1), we introduce $a[n]$ based on Dinkelbach optimization [19] to transform (P1) into

$$(P2) : \max_{\{r_i[n], \tau_i[n], l_{loc,i}[n]\}} \sum_{i=1}^I (l_i[n] - a[n] E_i[n]) \quad (20)$$

$$s.t. (4), (8), (9), (10), (16), (17), (18).$$

However, constraint (8) is binary, constraints (16), (17) and (18) are non-convex and the objective is non-convex, (P2) is difficult to solve.

We introduce auxiliary variables $\{t, t_{i,s}[n], t_{i,e}[n]\}$ to transform (P2) as

$$(P3) : \max_{\{r_i[n], \tau_i[n], l_{loc,i}[n]\}} t \quad (21a)$$

$$s.t. (4), (8), (9),$$

$$t \leq \sum_{i=1}^I (r_i[n] l_{off,i}[n] + (1 - r_i[n]) l_{loc,i}[n]) - a[n] \sum_{i=1}^I (E_i[n]), \quad (21b)$$

$$t_{i,s}[n] \leq \tau_i[n] \log_2 \left(1 + \frac{p_i[n] |h_{i,s}[n]|^2}{\sigma^2} \right), \quad (21c)$$

$$t_{i,e}[n] \geq \tau_i[n] \log_2 \left(1 + \frac{p_i[n] |h_{i,e}[n]|^2}{P_j |h_{j,e}[n]|^2 + \sigma^2} \right), \quad (21d)$$

$$c_s l_{off,i}[n] \leq F_s^{\max} \tau_i[n] \delta_t, \quad (21e)$$

$$r_i[n] l_{off,i}[n] + (1 - r_i[n]) l_{loc,i}[n] \geq q_{min}, \quad (21f)$$

$$r_i[n] l_{off,i}[n] + (1 - r_i[n]) l_{loc,i}[n] \leq q_i[n], \quad (21g)$$

$$q_i[n] + A_i[n] - (r_i[n] l_{off,i}[n] + (1 - r_i[n]) l_{loc,i}[n]) \leq Q_{max}. \quad (21h)$$

where $l_{off,i}[n] = B \delta_t (t_{i,s}[n] - t_{i,e}[n])$. t represents the calculating bits by TU_{*i*} offloading the information to SV. $t_{i,s}[n]$ and $t_{i,e}[n]$ are introduced to express the objective's lower bound, the data offloading rate's lower bound and the eavesdropping upper bound, respectively.

Problem (P3) is difficult to solve directly because (P3) is still non-convex. (P3) is divided into two parts, offloading decision and resource allocation. We use DRL to generate offloading

decision, where the DNN learns the offloading decision from the replay memory, and the resource is allocated by SCA [20].

A. Offloading decision Generation

In the Dinkelbach-guided DRL scheme, with given the feasible region of the resource allocation, the offloading decision generation problem is reformulated as

$$(P4) : (r^*[n]) = \arg \max_{\{r[n] \in \{0,1\}^I\}} G(r[n], \pi[n]), \quad (22)$$

where $\pi[n] = \{h[n], q[n]\}$ represents the input of DRL, which is consisted of $h[n] = \{\{h_{i,s}[n]\}_{i=1}^I, h_{j,e}[n], \{h_{i,e}[n]\}_{i=1}^I\}$ and $q[n] = \{q_i[n]\}_{i=1}^I$. $r[n] = \{r_i[n]\}_{i=1}^I$ represents the combination of all TUs offloading decisions. $G(r[n], \pi[n])$ represents the objective value with given $r[n]$ when the resource allocation problem is solved.

Note that the optimal $r^*[n]$ is obtained by substituting 2^I candidate offloading actions into (P3), which makes the complexity higher. Therefore, we aim to propose an efficient Dinkelbach-guided DRL scheme to highlight the security calculating efficiency performance of the UAV-aided MEC network with low complexity.

We design an online offloading decision function γ to generate offloading decision $r^*[n]$ in every frame through DNN, which learns the offloading decision from the replay memory,

$$\gamma : \pi[n] \rightarrow r^*[n]. \quad (23)$$

The parameters $\theta[n]$ of the DNN are initialized randomly with following standard normal distribution. We utilize the sigmoid function as the output function of the DNN. We obtain the slack offloading action of every TU $\hat{r}[n] = \{\hat{r}_1[n], \hat{r}_2[n], \dots, \hat{r}_I[n]\}$, which are all distributed between 0 and 1.

Then, we quantify the slack offloading action as M_n candidate actions through noisy-ordering-preserving (NOP) [17]. We first obtain $\frac{M_n}{2}$ candidate actions. The first one $s_1[n] = \{s_{1,1}[n], s_{1,2}[n], \dots, s_{1,I}[n]\}$ is obtained by the simple dichotomy,

$$s_{1,i} = \begin{cases} 0, & \hat{r}_i[n] < 0.5 \\ 1, & \hat{r}_i[n] \geq 0.5 \end{cases} \quad (24)$$

In order to get the $\frac{M_n}{2} - 1$ candidate actions, we sort the absolute values between $\hat{r}_k[n]$ and 0.5. The m_{th} one is obtained by

$$s_{m,i} = \begin{cases} 0, & \hat{r}_i[n] < \hat{r}_{m-1}[n], \\ 0, & \hat{r}_i[n] = \hat{r}_{m-1}[n] > 0.5 \cap \hat{r}_{m-1}[n], \\ 1, & \hat{r}_i[n] = \hat{r}_{m-1}[n] \leq 0.5 \cap \hat{r}_{m-1}[n], \\ 1, & [\hat{r}_i[n] > \hat{r}_{m-1}[n]]. \end{cases} \quad (25)$$

We obtain the remain candidate decisions by adding random noise σ' to the original slack offloading actions. Thus, the revised slack offloading action $\tilde{r}[n]$ is represented by

$\text{sigmod}(r[n] + \sigma')$. We can replace $r[n]$ with $\tilde{r}[n]$ and substitute $\tilde{r}[n]$ into (24) and (25) to obtain the remain candidate decisions.

Next, through optimizing resource allocation in section IV.C, we select the best offloading action $r^*[n]$ from M_n candidate actions,

$$r^*[n] = \arg \max_{r[n] \in \{s_m[n]\}_{m=1}^{M_n}} G(r[n], \pi[n]). \quad (26)$$

Then, we put $(\pi[n], r^*[n])$ into the setting replay memory, which has the limited size. In frame n , we randomly select training samples to update the DNN's parameters $\theta[n]$ if $\text{mod}(n, \omega_T) = 0$, where ω_T represents the training interval. The loss is given by

$$\text{Loss}(\theta[n]) = \frac{-1}{|\chi[n]|} \sum_{i \in \chi[n]} \left((r[i])^T \log f_{\theta[i]}(\pi[i]) + (1 - r[i])^T \log (1 - f_{\theta[i]}(\pi[i])) \right), \quad (27)$$

where $|\chi[n]|$ is the training sample's size.

B. Resource Allocation

With given TU_{*i*}'s offloading decision $r_i[n]$, the resource allocation problem of the UAV-aided MEC network is reformulated as

$$(P5) : \max_{\{\tau_i[n], t_{loc,i}[n], t_{i,s}[n], t_{i,e}[n]\}} t \quad (28)$$

s.t. (4), (9), (21b) – (21h).

Problem (P5) is convex due to all of its constraints are convex. (P5) can be easily solved by CVX [21]. Then, we can obtain the optimizing time factor and local calculating bits of each TU.

Then, the offloading decision and the resource allocation are optimized in frame n through the Dinkelbach-guided DRL scheme. The queueing module is obtained by doing the joint actions,

$$\pi[n+1] = \{h[n+1], q_i[n+1]\}_{i=1}^I. \quad (29)$$

We summarize the details of Dinkelbach-guided DRL algorithm for solving (P1) in algorithm 1.

V. SIMULATION RESULTS

In this section, simulation results are shown to prove the security calculating performance of our proposed Dinkelbach-guided DRL scheme. There are 5 TUs are randomly placed in the $400 * 400m^2$ area. The altitudes of SV and EV are both 100m. In horizontal coordinates, SV and EV fly in circles centered at (0,0) and radius $\rho_s = 70m$, $\rho_e = 60m$, respectively. One frame is $\delta_t = 0.5s$, Gaussian noise power is $\sigma = -110dBm$, Bandwidth is $B = 1MHz$. The maximum CPU frequency of TUs and SV $F_i^{max} = 1GHz$, $F_s^{max} = 10GHz$. The building DNN is consisted of one input, two hidden layers and one output.

Fig. 2 shows the security calculating efficiency of the UAV-aided MEC network versus time frames with different ρ_s .

Algorithm 1: Dinkelbach-guided DRL algorithm

Input: Channel coefficient $\{h[n]\}$, total frames N , number of candidate actions M_n , training interval ω_T .

Output: Joint offloading decision, time factor and local calculating bits $\{\{r_i[n]\}_{i=1}^I, \{\tau_i[n]\}_{i=1}^I, \{l_{loc,i}[n]\}_{i=1}^I\}$.

- 1 Initialize: TU_i initial data queue $\{q_i[0]\}_{i=1}^I$, the DNN parameters θ_1 and the replay memory;
- 2 **for** $n = 1, 2, \dots, N$ **do**
- 3 Input $\pi[n] = \{h[n], q_i[n]\}_{i=1}^I$;
- 4 Generate the slack offloading decision $\hat{r}[n] = \gamma(\pi[n])$ with the DNN;
- 5 Quantize $\hat{r}[n]$ into M_n binary actions by using NOP;
- 6 Optimize resource allocation to compute $G(r_m[n], \pi[n])$;
- 7 Select the best action $r^*[n] = \arg \max_{r_m[n]} G(r_m[n], \pi[n])$ and execute the joint optimizing action $(\gamma^*[n], y[n])$;
- 8 Update the replay memory through adding the joint actions $\{\{r_i[n]\}_{i=1}^I, \{\tau_i[n]\}_{i=1}^I, \{l_{loc,i}[n]\}_{i=1}^I\}$;
- 9 **if** $\text{mod}(n, \omega_T) = 0$ **then**
- 10 Randomly select training samples from the memory to update θ_n ;
- 11 **end**
- 12 $n = n + 1$;
- 13 Update $\{q_i[n]\}_{i=1}^I$ according to the joint actions $\{r^*[n-1], \tau_i[n-1], l_{loc,i}[n-1]\}$ and $\{A_i[n-1]\}_{i=1}^I$.
- 14 **end**

We can observe from Fig. 2 that the security calculating efficiency gradually converges with the larger frame. Because the parameters $\theta[n]$ of the DNN are trained to be more suitable for the dynamic communication environment with the time frames increase. We also find that the security calculating efficiency variation is different with different flying radius ρ_s . Because the trajectory of SV will make the communication channel coefficient different.

Fig. 3 shows the average security calculating efficiency of the UAV-aided MEC network versus $A_i[n]$ with different $p_i[n]$. We can observe that the average security calculating efficiency increases with the larger $A_i[n]$. Because TUs can offload more data to SV for calculation. We also find that the average security calculating efficiency is larger with the increase of $p_i[n]$. Because TUs can have more energy to offloading data with the increase of $p_i[n]$.

Fig. 4 shows the average security calculating efficiency variation of the UAV-aided MEC network with different CPU cycles for TU_i calculate one bit c_i . We can observe that the average security calculating efficiency is larger with the decrease of c_i . Because the calculating speed of TUs will be highlighted when c_i decreases. Then, the average amount of

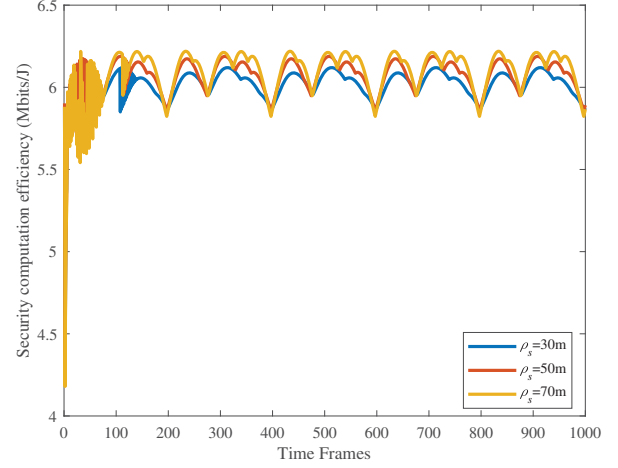


Fig. 2. Security calculating efficiency variation versus time frames with different flying radius ρ_s .

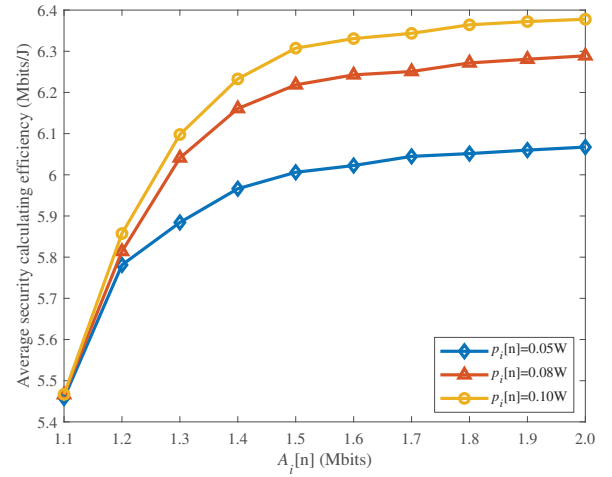


Fig. 3. Average security calculating efficiency variation versus $A_i[n]$ with different $p_i[n]$.

security calculating bits of the UAV-aided MEC network will increase.

In order to more intuitively reflect the efficient calculation of every TU's data information in the UAV-aided MEC network by our proposed Dinkelbach-guided DRL scheme, Fig. 5 shows the changes of TUs' data queue versus time frames with the initial data queue setting $2GMbits$ and the arrival data queue $A_i[n] = 0.5Mbits$. We can observe that data queue of all TUs are decreasing with the time frame increases even if the new data queues are arrived per frame. This proves the Dinkelbach-guided DRL scheme can efficiently assist TUs in calculating the data information in the UAV-aided MEC network and improve the security calculating efficiency performance.

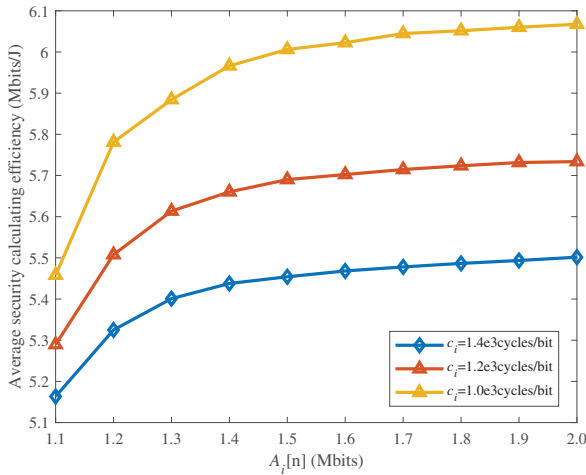


Fig. 4. Average security calculating efficiency variation versus $A_i[n]$ with different c_i .

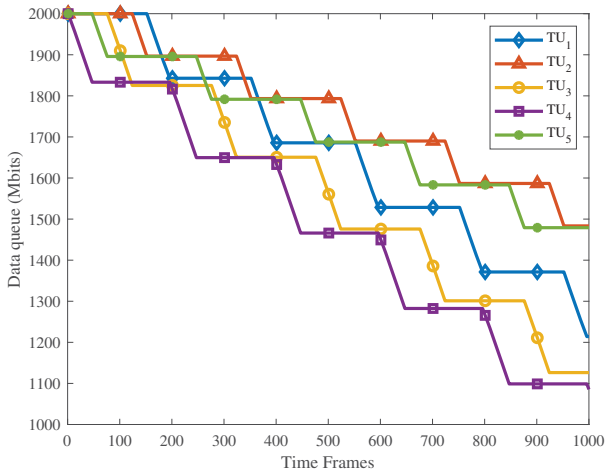


Fig. 5. Data queue variation of TUs.

VI. CONCLUSION

We propose a Dinkelbach-guided DRL scheme for secure communication in the UAV-aided MEC network. Specifically, the security calculating efficiency of the network is maximized by optimizing offloading decision and resource allocation under the condition of the data queue stability and minimum calculating requirement. The problem is intractable due to the fractional structure and binary constraint. Firstly, we deal with the fractional structure by taking advantage of Dinkelbach optimization. Then, we use DRL to generate offloading decision, where the DNN learns the offloading decision from the replay memory, and the resource is allocated by SCA. Simulation results show that the proposed Dinkelbach-guided DRL scheme efficiently improves the security calculating efficiency of the network.

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